groundplane

HDMI to MIPI CSI (RK628H) Board

RK628H 22P Board RevA.1
11/12/2026 9:34 AM

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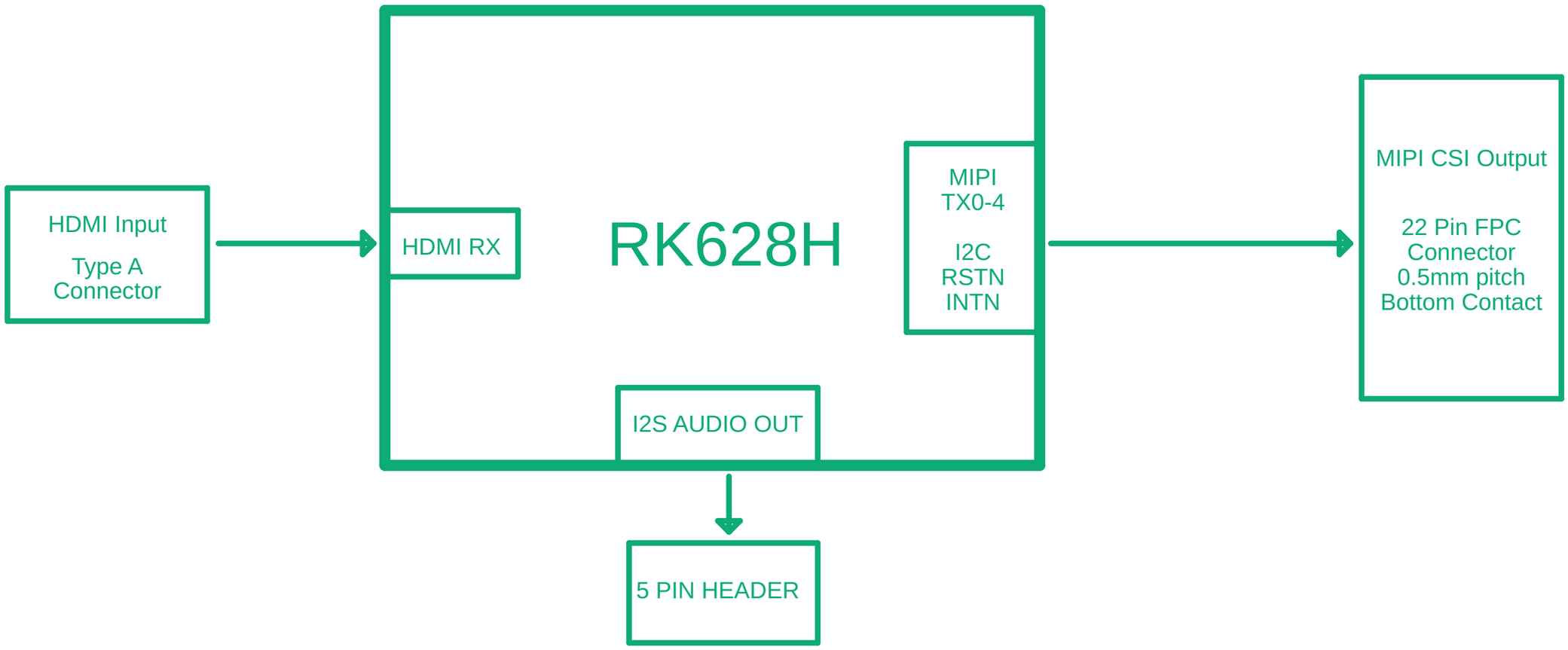
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BLOCK DIAGRAM

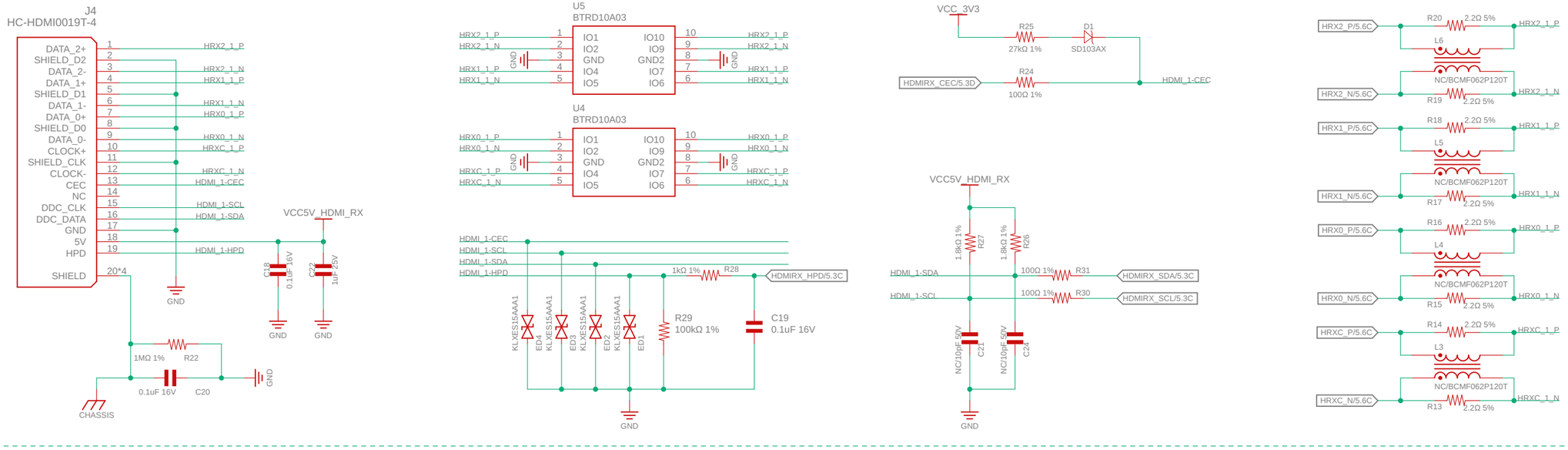
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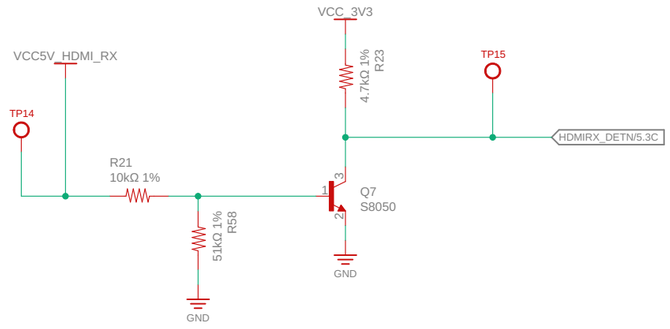


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HDMI INPUT





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HDMI AUDIO

U3
RK628H

FOR RK628H NO GVI

RK628_XTALI
RK628_XTALO
Y5
24MHz 20pF
C13 33pF 50V
C15 33pF 50V
R6 22Ω 1%

VCC_3V3
R1 NC/2.2kΩ 1%
R48 NC/2.2kΩ 1%
R2 2.2kΩ 1%
R56 10kΩ 1%
TP11
RK628_RSTN_3V3/6.7D
C5 0.1uF 16V
TP4 TP12 TP13
I2C_SCL_3V3/6.7D
I2C_SDA_3V3/6.7D
RK628_INTN_3V3/6.7D
R12 10kΩ 1%

TP1 TP2 TP3 UART Access for debug
R35 100Ω 1%

I2S_HDMI_SCLK/6.7B
I2S_HDMI_LRCK/6.7B
I2S_HDMI_SDI/6.7B
I2S_HDMI_MCLK/6.7B
R40 33Ω 1%
R41 33Ω 1%
R42 33Ω 1%
R43 33Ω 1%

HDMIRX_DETN/4.6E

HDMIRX_HPD/4.5C
HDMIRX_SDA/4.6C
HDMIRX_SCL/4.6C
HDMIRX_CEC/4.6B
R32 22Ω 1%
R37 22Ω 1%
R36 22Ω 1%
R33 22Ω 1%

R34 2kΩ 1%
HRXC_N/4.8D
HRXC_P/4.8D
HRX0_N/4.8C
HRX0_P/4.8C
HRX1_N/4.8C
HRX1_P/4.8C
HRX2_N/4.8B
HRX2_P/4.8B

R11 4.02kΩ 1%
MIPI-CSI0-D0_N/6.7E
MIPI-CSI0-D0_P/6.7E
MIPI-CSI0-D1_N/6.7E
MIPI-CSI0-D1_P/6.7E
MIPI-CSI0-CK_N/6.7D
MIPI-CSI0-CK_P/6.7D
MIPI-CSI0-D2_N/6.7D
MIPI-CSI0-D2_P/6.7D
MIPI-CSI0-D3_N/6.7D
MIPI-CSI0-D3_P/6.7D

C3/D5/E4
C26 10uF 10V
C29 0.1uF 16V
CH1 0.1uF 16V

J10
C23 10uF 10V
C25 0.1uF 16V

D4/E3/C5/F3
C33 10uF 10V
C34 0.1uF 16V
C31 0.1uF 16V
C32 0.1uF 16V

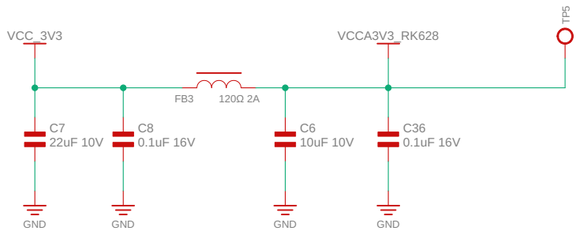
H9/J9
C28 10uF 10V
C27 0.1uF 16V

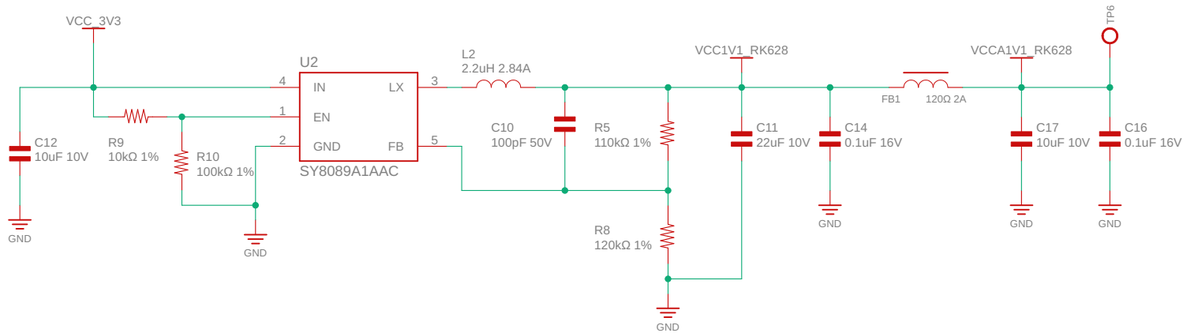
D8/E9
C30 0.1uF 16V

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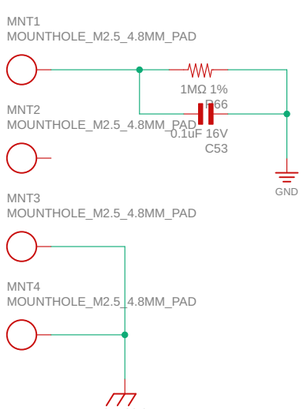
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POWER

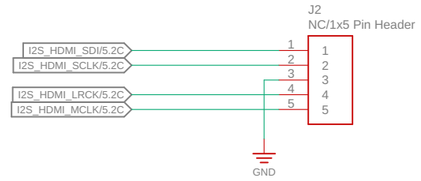




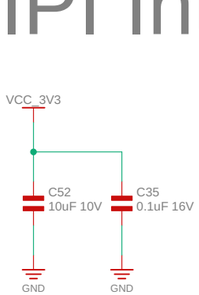
MOUNTHOLES

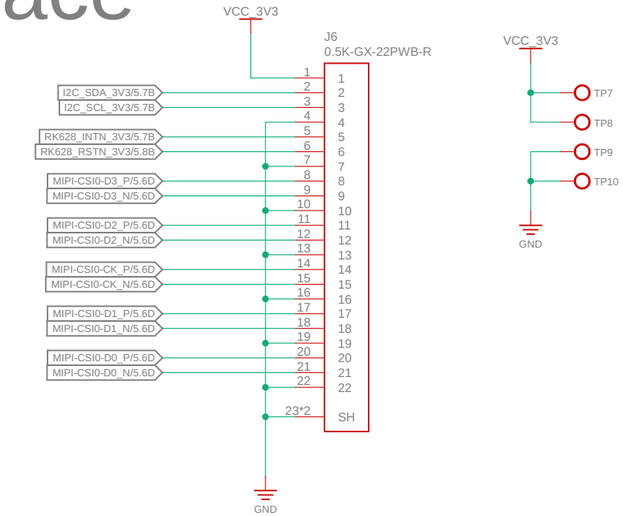


I2S Audio Header



MIPI Interface





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